# Session 2 (FIND & READ): Solutions and facilitation notes

**Foundations of Clinical Research · Level 1** · Neudata · *Instructor copy. Do not share before the session.*

Model answers. Other answers are fine if they are specific and consistent.

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## Recap quiz: Session 1 answers

1. **Paper 1 (Boulet 2024) is a trial** (pilot RCT); **Paper 2 (Hyun 2023) is a cohort** (matched registry cohort).
2. **P** adults with sepsis · **E** positive cumulative fluid balance by day 3 · **C** neutral or negative balance · **O** death · **T** by day 28 (a PECO question: E = Exposure).
3. **Feasible**: a mortality outcome needs more than 1,000 patients.
4. **"To estimate the proportion fluid-positive on day 3"**: a measurable verb. "Understand" is not measurable.
5. **H0:** cumulative fluid balance at day 5 is the same with the restrictive protocol and with usual care.

If many miss 4 or 5, spend one extra minute: objectives use measurable verbs; H0 always states "no difference".

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## Check your understanding (segment 2.1): AND, OR or NOT?

1. **AND** (septic shock AND fluid restriction). 2. **OR** (restrictive OR conservative). 3. **Fewer** (AND narrows). 4. It removes every paper that mentions children anywhere, including good adult studies that mention children once.

## Check your understanding (segment 2.2): OK or not OK?

1. **Not OK**: patchwriting is plagiarism even with a citation. 2. **OK** (but rare in medical writing). 3. **Not OK**: someone else's finding needs a citation. 4. **Not OK**: you may repeat an error, so read and cite the original.

## Check your understanding (segment 2.3): spot the spin

1. Causal word for an observational design. Rewrite as *"a positive fluid balance was associated with higher mortality"*.
2. p = 0.31 is no evidence of a difference. Rewrite as *"no significant difference; the confidence interval was wide"*.
3. Headline is a secondary outcome. Lead with the primary result instead.
4. Before–after change without a comparison group. Rewrite as *"mortality was lower after implementation; without a control group we cannot attribute this to the protocol"*.

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# Exercise 1: Boolean search builder

**Warm-up:** `septic shock OR fluids` gives more hits (either term). `sepsis` gives more hits than `sepsis NOT neonatal` (NOT removes papers).

**Model string for the running example** (as in the demo):

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab])
AND ("Fluid Therapy"[Mesh] OR "fluid restrict*"[tiab] OR "restrictive fluid*"[tiab]
  OR "conservative fluid*"[tiab])
AND ("Mortality"[Mesh] OR mortality[tiab] OR death[tiab])
```

Sensible extra synonyms for the I block: `de-resuscitation[tiab]`, `deresuscitation[tiab]`, `"fluid balance"[tiab]`, `"Water-Electrolyte Balance"[Mesh]`.

Sensible filters (each with a reason): article type *Randomized Controlled Trial* + *Systematic Review* (treatment question); *Adult: 19+ years* (our population); publication date *10 years* (current practice). Avoid *Free full text* (loses key papers).

**Model string for Scenario C (delirium)**

```
(Delirium[Mesh] OR delirium[tiab] OR "acute confusion"[tiab])
AND ("Hip Fractures"[Mesh] OR "hip fracture*"[tiab])
AND (prevent*[tiab] OR "multicomponent"[tiab] OR "non-pharmacological"[tiab] OR
  bundle[tiab])
```

**Common faults**

- Missing brackets: `A OR B AND C` is read differently from `(A OR B) AND C`.
- AND inside a block (`septic shock AND sepsis`). Synonyms need OR.
- A block for the comparison ("usual care"). It adds nothing and loses papers.
- `NOT` used to remove a topic the paper only mentions in passing.
- Search words in the local language: PubMed indexes English titles and abstracts.

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# Exercise 2: Paraphrasing (model answers)

**Part A**

| Version | Verdict | Why |
|---|---|---|
| a) | **Problem: patchwriting** | Same structure and almost the same words as the source; one word swapped ("mortality" for "death"). Plagiarism even though it is cited. |
| b) | **OK** | New structure, adds context (matched registry cohort, 28-day mortality), keeps the calibrated word "associated", gives numbers and the citation. |
| c) | **Problem: distorts the source** | "Giving more fluid ... doubles the risk" turns an observational *association* into causation, and "doubles" overstates (HR 2.13 only for day 2; 1.56 for day 3). Citation does not fix misrepresentation. |

| Version | Verdict | Why |
|---|---|---|
| d) | **Problem: no citation** | Good wording, but presenting someone else's finding without a citation is plagiarism. |

**Part B: model paraphrase**

"Mortality and days alive without organ support at day 28 did not differ between the two groups, although the pilot trial was too small to detect such differences (1)."

What good looks like: new structure; facts unchanged (28 days, mortality, days alive without life support, similar); no stronger claim (do **not** write "restriction is safe"); citation. Adding "the trial was too small" is accurate context from the paper (48 patients; not powered for mortality).

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## Exercise 3: Paper 1's abstract (answer key)

| Item | Answer (from the abstract) |
|---|---|
| Design (and blinding) | Multicentre, randomised, controlled **pilot** trial; single-blind (patients and statisticians blinded, clinicians not) |
| P | Adults 18–85 years with septic shock, in the first 24 h of vasopressor infusion, in three ICUs; excluded: AKI needing renal replacement, end-stage CKD, severe malnutrition |
| I vs C | Optimised fluid restriction (reducing maintenance fluids and fluids for drug dilution during the first 7 days) vs standard fluid strategy |
| Primary outcome | **Cumulative fluid balance (ml/kg) in the first 5 days**, not mortality |
| Numbers | 1,201 screened; 50 randomised; 2 withdrew (control); 48 analysed (24 per group) |
| Main result | Median cumulative balance at day 5: 6.9 vs 35.0 ml/kg; absolute difference 13.2 ml/kg (95% CI −15.2 to 41.6), p = 0.42. |
| Conclusion | "...did not reduce the overall fluid balance at day 5" |
| Matches the result? | **Yes**. The CI includes zero, and the conclusion says "did not reduce". |
| Spin in the abstract? | **No.** The abstract conclusion matches the primary result. The spin is in the *Discussion*, which calls the result a "non-significant trend" towards a lower fluid balance. With p = 0.42 and a 95% CI from −15.2 to 41.6 ml/kg, the data fit no difference at all, so the word "trend" suggests an effect the trial did not show. |
| Registration | ClinicalTrials.gov **NCT04947904** (registered 1 July 2021, before recruitment began in September 2021) |
| Answers our 28-day mortality question? | **No.** It is a small pilot powered for fluid balance. 28-day mortality was a secondary outcome and was similar between groups: "no evidence of a difference", not "no difference". |

**Fast finishers (Paper 2):** the title says **"Impact"** (causal); the abstract conclusion correctly says **"associated"**.

**Common errors to correct in the debrief**

- Writing "mortality" as the primary outcome.
- Copying only the p-value and not the estimate with its 95% CI.
- Calling the trial "double-blind". Clinicians, who prescribe the fluids, were not blinded.
- Reading the medians (6.9 vs 35.0) as "five times less fluid balance, so it works". The CI crosses zero.

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## Exercise 4: Proposal builder 2 (what good looks like)

The trainer models this live for the running example (the feasible pilot version A from Session 1); each participant then does the same for their own question. Use it as the quality benchmark, not as the answer.

**Search (documented):** PubMed, date, the string from Exercise 1, filters (article type RCT + systematic review; adults; 10 years), number of hits; plus a check of ClinicalTrials.gov for ongoing trials.

**Working title:** *Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial* (13 words).

**Keywords:** Fluid Therapy; Water-Electrolyte Balance; Critical Care; Vasoconstrictor Agents (MeSH terms not already in the title).

**Problem statement (≈ 300 words, three paragraphs)**

1. *Known:* Septic shock is a leading cause of death in the intensive care unit (ICU). Early fluids restore perfusion, but fluids given after the first hours often lead to a positive fluid balance, which is associated with higher mortality in observational studies (1).
2. *Not known:* Two large randomised trials found that restricting fluids did not change mortality (2,3). A recent pilot trial that targeted non-resuscitation fluids did not significantly reduce fluid balance at day 5 (4). It therefore remains unclear whether fluid balance can be reduced reliably in routine care.
3. *Aim and objectives:* In this pilot randomised trial, we aim to assess whether a restrictive fluid protocol reduces cumulative fluid balance at day 5 compared with usual care in adults with septic shock. Our primary objective is to compare day-5 fluid balance between the two groups; our secondary objective is to assess the feasibility of a larger trial.

References, in the order first cited: (1) Hyun 2023; (2) CLASSIC, Meyhoff 2022; (3) CLOVERS, Shapiro 2023; (4) Boulet 2024.

*Science rule:* the Introduction never contains your own data or results (no "our audit"). They belong in the Results. A local figure may appear only if it comes from a publication, with its reference.

**Five key references (model)**

| # | Reference | Why included |
|---|---|---|
| 1 | Boulet N, et al. *Crit Care* 2024;28:429 (pilot RCT, n = 48) | Closest design and outcome; primary outcome not significantly different |

| # | Reference | Why included |
|---|---|---|
| 2 | Hyun DG, et al. *Ann Intensive Care* 2023;13:105 (matched cohort) | Positive balance on days 2–3 associated with higher 28-day mortality |
| 3 | Meyhoff TS, et al. (CLASSIC) *N Engl J Med* 2022;386:2459–70 | Large RCT, septic shock, no difference in 90-day mortality |
| 4 | Shapiro NI, et al. (CLOVERS) *N Engl J Med* 2023;388:499–510 | Large RCT, sepsis-induced hypotension, restrictive vs liberal, no difference in 90-day death before discharge |
| 5 | Evans L, et al. Surviving Sepsis Campaign guidelines 2021. *Crit Care Med* 2021;49:e1063–e1143 | Current guideline; shows the remaining uncertainty |

*Instructor note:* check citation details against PubMed before distributing. Participants must verify every reference themselves, which is exactly the habit to practise.

**The live part (minutes 0–10): what the trainer shows**

1. Run the full string from the PubMed demo (`s2_demo.md`, Strings to copy, Step 3) with the filters RCT and systematic review, adults, 10 years. Write the search record on screen: database, date, string, filters, number of results. The number changes every day, so read it from the screen.
2. Screen the first titles aloud. Keep adults with sepsis or septic shock, a comparison of fluid strategies or fluid balance, and a trial, cohort or review. Skip children, burns and surgery, and trials of one fluid type against another (crystalloid against colloid). Participants suggest papers and keywords in the chat.
3. Save Boulet 2024 and CLASSIC to Zotero (*Send to → Citation manager*) and fill their two worksheet rows together (model rows below).
4. Search ClinicalTrials.gov for "septic shock" and "fluid" to show how to look for ongoing trials.
5. Ask for keywords in the chat. Model answer: the four MeSH terms above.

**Literature review worksheet: model rows**

| # | Study | Design, size | Compared | Primary outcome: result | Key limitation | Why it matters for our question |
|---|---|---|---|---|---|---|
| 1 | Boulet 2024, *Crit Care* | Pilot RCT, 3 ICUs, 50 randomised, 48 analysed | Restricting maintenance and drug-dilution fluids for 7 days vs standard care | Cumulative fluid balance at day 5: median 6.9 vs 35.0 ml/kg, p = 0.42, no significant difference | Small, not powered for mortality, clinicians not blinded | Closest design and outcome to our pilot |
| 2 | Hyun 2023, *Ann Intensive Care* | Propensity-matched cohort, multicentre registry, 20 hospitals, 2,516 patients | Positive vs negative cumulative fluid balance at the end of ICU day 1, 2 or 3 | 28-day death: HR 1.17 (95% CI 0.85 to 1.60) at day 1, 2.13 (1.48 to 3.06) at day 2, 1.56 (1.10 to 2.22) at day 3 | Observational, confounding possible | Fluid balance is linked with death, so it is worth reducing |

| # | Study | Design, size | Compared | Primary outcome: result | Key limitation | Why it matters for our question |
|---|---|---|---|---|---|---|
| 3 | Meyhoff 2022, *N Engl J Med* (CLASSIC) | RCT, 1,554 adults with septic shock | Restrictive vs standard intravenous fluid | 90-day death 42.3% vs 42.1%, no difference | Clinicians not blinded | Restriction did not change mortality in a large trial |
| 4 | Shapiro 2023, *N Engl J Med* (CLOVERS) | RCT, 1,563 adults with sepsis-induced hypotension | Restrictive (earlier vasopressors) vs liberal fluid | Death before discharge home by day 90: 14.0% vs 14.9%, no difference | Open label, early phase of care only | Confirms CLASSIC in a different population |
| 5 | Evans 2021, *Crit Care Med* (Surviving Sepsis Campaign) | International guideline | Not applicable | No recommendation on restrictive vs liberal fluid after initial resuscitation (insufficient evidence) | Expert judgement where evidence is weak | Shows the uncertainty our pilot addresses |

### Feedback guide when reading drafts (not a grade)

| Element | Good | Needs work |
|---|---|---|
| Search | String, database, date, filters and hits recorded; brackets correct | "We googled it"; no date; AND inside a block |
| References | 5, relevant, include the best available design, exported to a reference manager | Websites, textbooks only, typed by hand, unread |
| Literature review worksheet | Same columns for every paper; limitation column filled | Copied abstract sentences |
| Problem statement | Known, then gap, then question and objectives; each fact cited | A literature dump; own data or results; unpublished local figures |
| Title & keywords | 10–16 words; P, I/E, C and design visible; calibrated words; keywords add MeSH not in the title | "Fluids in the ICU"; abbreviations; "impact of" for an observational design |
| Integrity | Own words, every claim cited | Sentences pasted from abstracts |

## Exit question: answers

1. **OR** gives more hits.
2. Paper 1's primary outcome was **cumulative fluid balance at day 5 (ml/kg)**, not mortality.